1. Berlin definition and severity

Criteria

- Within 1 week of insult; bilateral opacities not fully explained by effusion/collapse
- Not fully explained by cardiac failure/overload (echo if no risk factor)
- PaO2/FiO2 with PEEP ≥ 5: **mild 200–300, moderate 100–200, severe ≤ 100**

2023 global definition

- Includes HFNO ≥ 30 L/min, SpO2/FiO2 ≤ 315, lung ultrasound in resource-limited settings

Key point: Severe ARDS = P/F ≤ 100: say it, then escalate in steps.

2. Lung-protective ventilation

Core

- **Vt 6 mL/kg predicted body weight** (ARDSNet ARMA 2000: mortality 40 → 31%)
- PBW male = 50 + 0.91 × (height cm − 152.4); female 45.5 + …
- **Plateau ≤ 30 cmH2O**; **driving pressure ≤ 15** (Amato 2015: strongest predictor)
- Permissive hypercapnia, pH ≥ 7.20–7.25
- SpO2 88–95% / PaO2 7.3–10.7 kPa

Mechanical power

- Energy per minute (Vt, ΔP, RR, flow): lower is safer; avoid high RR

Key point: 6 mL/kg PBW, Pplat ≤ 30, ΔP ≤ 15.

3. PEEP and recruitment

PEEP

- Higher PEEP in moderate-severe ARDS (ALVEOLI, LOVS, EXPRESS: meta-analysis benefit if P/F < 200)
- Methods: PEEP/FiO2 table, best compliance, oesophageal pressure (EPVent-2: neutral), EIT
- Check recruitability: P/F and compliance response

Recruitment manoeuvres

- **ART 2017: staircase RM + titrated PEEP ↑ mortality** → avoid prolonged high-pressure RM
- Short RM only after disconnection/derecruitment

Key point: Individualise PEEP; avoid routine aggressive recruitment (ART).

4. Prone positioning

Evidence

- **PROSEVA 2013**: P/F < 150 (FiO2 ≥ 0.6, PEEP ≥ 5), early (< 36 h), ≥ 16 h/day → 28-day mortality 33 → 16%

Practice

- Trained team, pressure areas, ETT/line safety, eyes, enteral feed
- Stop when P/F ≥ 150 with PEEP ≤ 10, FiO2 ≤ 0.6 for 4 h after supine
- Contraindications: unstable spine, open abdomen, raised ICP (relative), pregnancy (relative, feasible)

Key point: Prone early and long: PROSEVA is the single most important ARDS trial to quote.

5. Neuromuscular blockade and inhaled vasodilators

NMB

- ACURASYS 2010: cisatracurium 48 h ↓ mortality; **ROSE 2019: no benefit** with lighter sedation and higher PEEP
- Use for dyssynchrony, high Pplat despite sedation, during proning

Inhaled NO / prostacyclin

- Improve oxygenation transiently; **no survival benefit**, ↑ AKI with NO
- Rescue bridge to prone/ECMO, or RV failure

Steroids

- Dexamethasone (DEXA-ARDS) for moderate-severe ARDS; COVID-19 (RECOVERY); CAP (CAPE COD hydrocortisone)

Key point: Oxygenation improvement does not mean survival benefit.

6. V-V ECMO

Referral criteria (EOLIA)

- P/F < 50 for > 3 h, or < 80 for > 6 h
- pH < 7.25 with $PaCO_2$ ≥ 8 kPa for > 6 h (RR 35, Pplat ≤ 32)
- Despite optimal care incl. prone and NMB

Evidence

- CESAR 2009: referral to ECMO centre ↓ death/disability
- EOLIA 2018: 35% vs 46% (NS), but crossover 28%; Bayesian and meta-analysis favour ECMO

Contraindications

- Ventilated > 7 days with high settings (relative), irreversible disease, severe immunosuppression, contraindication to anticoagulation

Key point: Refer early to a high-volume centre; ECMO is a bridge, not a cure.

7. Prognosis and ethics

Mortality predictors

- Age, severity (P/F), driving pressure, dead space fraction, non-pulmonary organ failure, immunosuppression
- RESP score (ECMO survival), PRESERVE score

Ethics

- Resource allocation, candidacy, bridge to nowhere, family communication, time-limited trial, withdrawal on ECMO

Key point: Finish the answer with prognosis and an ethical sentence.

8. Non-invasive support and specific causes

HFNO / NIV

- FLORALI: HFNO ↓ intubation in P/F ≤ 200 and ↓ mortality vs NIV/O2
- Watch HACOR score / ROX index; avoid delayed intubation and P-SILI

Cause-specific

- CAP: early antibiotics, steroids (CAPE COD) in severe CAP
- Influenza: oseltamivir; COVID-19: dexamethasone, IL-6 blockade
- Conservative fluid strategy once shock resolved (FACTT)

Key point: Treat the cause while protecting the lung.